

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

Class	Section Number
8:50	001
9:55	004

As you complete the exam, write your initials at the top right of each other page.

When the exam start time is called, you may turn the page and begin your exam.

If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

Some multiple choice questions are "Select ONE" while others are "Select ALL that apply". Pay attention to the question type and only mark one option if it says "Select ONE". Fill in the circles completely. Only highlight the circles of the selected options and leave the others completely blank.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

1. Some of the following problems are based on the `baseball` dataset, which records batting, pitching, and fielding statistics for 50 players in a community baseball league. Each player, identified by `id`, has batting statistics recorded. The `position` column identifies whether the player has fielding (FIE) or pitching (PIT) statistics recorded. The first six rows and a brief description of the statistics are given below.

id	BAT_RBI	BAT_AVG	position	FIE_DBP	FIE_ERR	PIT_AVG	PIT_SHO
2	4	0.126	infield	53	5	NA	NA
35	4	0.124	pitcher	NA	NA	7.026	1
50	7	0.174	pitcher	NA	NA	7.166	0
3	8	0.250	infield	66	15	NA	NA
33	4	0.142	infield	91	9	NA	NA
19	4	0.128	outfield	59	14	NA	NA
.... plus 44 more rows							

- `BAT_RBI`: runs batted in
- `BAT_AVG`: overall batting average
- `FIE_DBP`: double-plays earned
- `FIE_ERR`: fielding errors made
- `PIT_AVG`: average strikeouts per game
- `PIT_SHO`: shutout games (no runs scored)

The following players are eligible to win an award from the baseball league:

- (A) Any player with a batting average above 0.28
- (B) Any pitcher with at least one shutout game (`PIT_SHO`)
- (C) Any non-pitcher (infield or outfield) with more than 70 double plays and fewer than 10 errors

Consider using R to identify these players in the data. For example, to return rows who meet condition (A) we can use:

```
baseball %>% filter(BAT_AVG > 0.28)
```

- (a) (4 points) What inputs can go into `filter()` to find rows that meet condition (B)? **Select ALL that apply.**

- ☒ `position == "pitcher" & PIT_SHO > 0`
☒ `position == "pitcher" & PIT_SHO != 0`
☐ `!is.na(PIT_SHO)`
☐ `position == "pitcher" | PIT_SHO > 0`

- (b) (4 points) What inputs can go into `filter()` to find rows that meet condition (C)? **Select ALL that apply.**

- ☐ `position %in% c("infield", "outfield") & (FIE_DBP > 70 | FIE_ERR < 10)`
☒ `position %in% c("infield", "outfield") & (FIE_DBP > 70 & FIE_ERR < 10)`
☒ `position != c("pitcher") & (FIE_DBP > 70 & FIE_ERR < 10)`
☐ `position != c("pitcher") | (FIE_DBP > 70 & FIE_ERR < 10)`

- (c) (4 points) The dataframe `past_awardees` contains the 11 players who have previously won an award, identified by the same `id` column as in `baseball`. How can the `baseball` data be joined with the `past_awardees` to ensure that the rows in the output are all players who have **not** received an award before?

You should identify **which** join function to use as well as the **order** of the two datasets, and briefly explain your answer.

Solution: We should use `anti_join(baseball, past_awardees)`. This will get us a filtered version of `baseball` where the output is all rows that do not appear in `past_awardees` (i.e. players who have not already won an award).

2. Your zoologist friend is studying three species of zebra. She has gathered data on the physical characteristics of 160 of these zebras and asks for your help analyzing it. Below are 5 example rows from full dataset.

	length	height	sex	species
	<dbl>	<dbl>	<chr>	<chr>
1	237	142	F	mountain
2	245	145	M	plains
3	216	128	F	mountain
4	235	139	F	plains
5	282	152	M	grevy's

She sends you the following message about how she wants to graph the data: "I want to make a scatter plot of each zebra's body length and shoulder height, with a single smooth line showing the trend. I want the color of the points to be different for the different species, and I want all of the points to be slightly transparent in case they overlap".

- (a) (6 points) Which ggplot2 graph types should you add to the plot to fulfill her request? **Select ALL that apply.**

- ☐ geom_density()
 ☒ geom_smooth()
- ☒ geom_point()
 ☐ geom_col()
- ☐ geom_line()
 ☐ geom_bar()

- (b) (4 points) For each of the following aesthetics in your plot, identify whether they would need to be global variable, local variable, or constant. **Select ONE each.**

- x : ☒ Global ☐ Local ☐ Constant
- y : ☒ Global ☐ Local ☐ Constant
- color : ☐ Global ☒ Local ☐ Constant
- alpha : ☐ Global ☐ Local ☒ Constant

- (c) (3 points) Your friend also wants to make side-by-side plots to compare the range of heights for the three different species and identify any potential outliers. Would a histogram or boxplot work better for this? Briefly explain your answer.

Solution: Side-by-side boxplots would work better since they show not only the range of each group but also calculate and mark outliers based on each group's own IQR. A histogram would not delineate outliers but it would give a better idea of the specific shape of each group.

3. (5 points) The Language Fluency dataset, saved in the environment as `Language`, contains student names, each parent's first language, and indicator variables for five languages. Each language column specifies whether the individual speaks that language (e.g., TRUE/FALSE).

```
A tibble: 50 × 8
  Name      Mom_speaks Dad_speaks English Chinese French Arabic Spanish
<chr>      <chr>      <chr>      <lgl>  <lgl>  <lgl>  <lgl>  <lgl>
Henry Todd  English    English    TRUE   FALSE  FALSE  FALSE  TRUE
Ren Xiao    Japanese  Chinese    TRUE   TRUE   FALSE  FALSE  FALSE
Wendy Li    Chinese    English    TRUE   FALSE  FALSE  FALSE  TRUE
Taylor Leight English    English    TRUE   FALSE  TRUE   FALSE  TRUE
Joseph Lopez Spanish    English    TRUE   FALSE  FALSE  FALSE  FALSE
Aaron Sato  English    Japanese   TRUE   TRUE   FALSE  FALSE  FALSE
.... plus 44 more rows
```

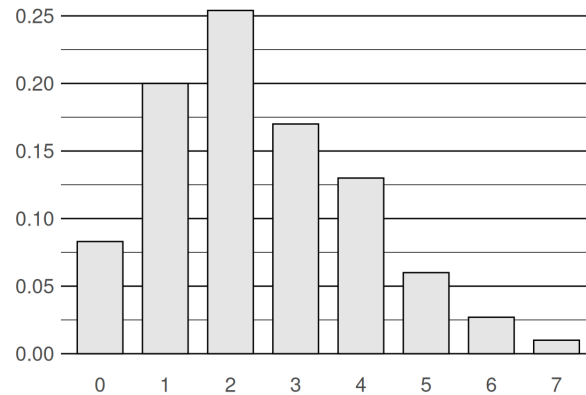
The *Undergraduate Major and GPA* dataset saved in the environment as `Major` contains the names of 20 students (15 of them are also present in the `Language` dataset), their Undergraduate majors, and other columns.

```
A tibble: 20 × 5
  Name      Undergrad_major      Year_of_graduation  GPA Grad_School
<chr>      <chr>                        <dbl> <dbl> <chr>
Henry Todd  Computer Science              2020  3.45 YES
Monica Keale Biology                      2021  2.78 NO
Wendy Li    Psychology                   2027  3.1  NA
Taylor Leight Business Administration  2022  3.92 YES
Ezequial McPherson English Literature          2024  3.88 NO
Joseph Lopez Mathematics              2026  3.98 YES
.... plus 14 more rows
```

Working with the `Language` and `Major` and predict the dimension of the output of the following joins. Assume that `join_by` is automatically set to join correctly by the student's name.

Join function	Rows	Columns
<code>semi_join(Language, Major)</code>	<u>15</u>	<u>8</u>
<code>full_join(Language, Major)</code>	<u>55</u>	<u>12</u>
<code>right_join(Language, Major)</code>	<u>20</u>	<u>12</u>
<code>right_join(Major, Language)</code>	<u>50</u>	<u>12</u>
<code>anti_join(Major, Language)</code>	<u>5</u>	<u>5</u>

4. The distribution of goals in a game in the FIFA World Cup is given in the graph below.



- (a) (3 points) Find the 25th, 50th and the 75th percentile of the number of goals.

Solution: The percentiles are based on the lower probabilities. 8% of the population is less than or equal to 0, about $8\% + 20\% = 28\%$ is ≤ 1 , about $28\% + 25\% = 53\%$ is ≤ 2 , about $53\% + 17\% = 70\%$ is ≤ 3 , and about $70\% + 13\% = 83\%$ is ≤ 4 .

So, the 25th percentile is 1, the 50th percentile is 2, and the 75th percentile is 4. These are the values of that capture at least that much probability below.

- (b) (3 points) Is the mean equal to, smaller than, or larger than the 50th percentile for the number of goals? Explain how you can tell without needing to know the exact value of the mean.

Solution: The mean must be larger than the 50th percentile (the median) because of the skew in the distribution. Most of the values are on the left side of the graph, around 1 or 2, but there is a tail of bigger values going to the right. The mean will be pulled higher because of this right tail.

- (c) (2 points) We want to re-create this graph with 'ggplot()'. Since we know the probabilities of each number 0-7, which 'geom' should we use? **Select ONE.**

☐ geom_bar()
☒ geom_col()

☐ geom_histogram()
☐ geom_density()

5. The dataset 'bm_sample' records the sex, age, and completion time for several runners of the Boston Marathon in both 2010 and 2011.

BibNum <dbl>	Year <dbl>	sex <chr>	Age <dbl>	age_range <chr>	total_time <dbl>
5668	2011	male	53	50-54	188.55
2863	2011	male	53	50-54	199.19
14490	2010	female	33	18-34	240.84
18743	2011	female	41	40-44	247.73
13823	2011	female	28	18-34	209.25
27033	2010	female	35	35-39	295.54

- (a) (4 points) Suppose the age_range column does not exist yet. Fill in the blanks below to identify which two dplyr commands to use to build the categorical age_range from the numeric Age.

```
bm_sample %>%
  mutate (age_range =
    case_when (Age < 35 ? "18-34",
               Age < 40 ? "35-39",
               Age < 45 ? "40-44",
               Age < 50 ? "45-49",
               Age < 55 ? "50-54",
               .default = "55 and older" ))
```

- (b) (2 points) What symbol should we use to assign categories in the code above? In other words, what symbol should replace the ?s?

Solution: Within `case_when()`, we use the `~` symbol when we define categories based on a condition.

- (c) (6 points) We want to create the following the summary table for all the male participants:

```
A tibble: 2 × 5
  Year   n_male avg_time sd_time n_total
<dbl> <int>   <dbl>   <dbl>   <dbl>
2010   2802   220.5967  37.76457  4940
2011   2942   222.8564  39.66779  5060
```

```
bm_sample %>%
  filter(sex == male) %>%
  group_by(sex) %>%
  summarise(n_male = count(),
            avg_time = mean(total_time),
            sd_time = sd(total_time)) %>%
  mutate(n_total = c(4940, 5060))
```

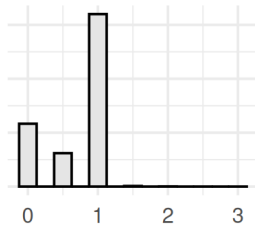
The code above is supposed to produce the summary table, but it does not work correctly. Identify the problems with the code and **Select ALL** fixes from the list below.

- ☒ Change `sex == male` to `sex == "male"` in `filter()`.
- ☐ Change `sex == male` to `sex = male` in `filter()`.
- ☐ Change `sex` to `"sex"` in `group_by()`.
- ☒ Change `sex` to `Year` in `group_by()`.
- ☒ Change `count()` to `n()` in `summarise()`.
- ☐ Change `summarise()` to `mutate()`.

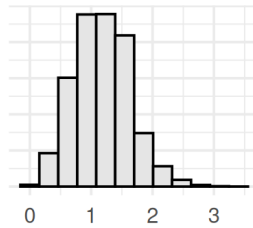
6. The number of muscle sprains sustained in a year is recorded for a sample of 95 college athletes. The data has range 0-6, median 1, mean 1.16, and has a long right tail.

(a) (4 points) The graphs below show several different sampling distributions. Match each graph to the description of a sampling distribution below.

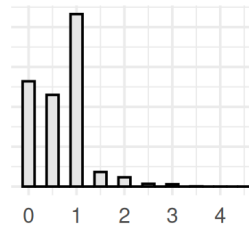
Graph A



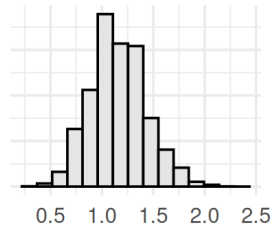
Graph B



Graph C



Graph D



- B The sample mean muscle sprains from a resample of 10 athletes.
 - D The sample mean muscle sprains from a resample of 30 athletes.
 - C The sample median muscle sprains from a resample of 10 athletes.
 - A The sample median muscle sprains from a resample of 30 athletes.
- (b) (3 points) Which of the following probabilities is largest? **Select ONE.**
- ☐ A single random athlete sustains 2 or fewer muscle sprains.
 - ☐ The average number of muscle sprains sustained by 10 random athletes is less than or equal to 2.
 - ☒ The average number of muscle sprains sustained by 30 random athletes is less than or equal to 2.
- (c) (4 points) A new sample of 85 athletes is taken a year after new safety policies are introduced, and the number of muscle sprains sustained is recorded for each one. The code below is used to resample from the original data (sprains_old) and the new data (sprains_new), and compare the difference in sample means between the two datasets.

```
new_data <- function(data, n) {
  sample_n(data, size = n, replace = TRUE)
}

resampled_means <- tibble(
  mean_old = replicate(10000,
    mean(new_data(sprains_old, 95)$nsprains) ),
  mean_new = replicate(10000,
    mean(new_data(sprains_new, 85)$nsprains) ),
  diff = mean_old - mean_new)
```

Explain why the resampling code above does **not** reflect a null hypothesis that the mean number of sprains is the same between the two groups? How would you change this code to incorporate this null hypothesis? (You do not need to re-write the code, a written explanation is sufficient.)

Solution: The code assumes that the old and new groups are drawn from completely different population distributions, which is why they are being resampled from each of the respective original data sets.

If we wanted to do a resampling experiment of the null hypothesis of equal means, we would need to do a “big box” resampling procedure by combining all of the data into one single large dataset. Then we would resample 95 rows for the “old” data and 85 rows from the “new” data, both from the big box. This reflects the assumption that both groups are drawn from the same underlying population.

7. An internet provider takes a sample of 200 customers and finds that 63% are satisfied with their service. A 99% confidence interval is created via resampling and is reported as

$$(0.545, 0.72)$$

- (a) (4 points) Write an expression for M_{lower} , the 0.5th percentile of the resampled proportions. You do not need to simplify your answer and can leave it as a numerical expression.

Solution: The formula for the resampled CI is $(2M_{obs} - M_{upper}, 2M_{obs} - M_{lower})$ which means

$$0.72 = 2M_{obs} - M_{lower}$$

$$0.72 = 2(0.63) - M_{lower}$$

$$M_{lower} = 2(0.63) - 0.72$$

- (b) (3 points) Suppose we had instead built a 95% confidence interval based on the same set of resamples. Which of the following statements is true? **Select ONE.**

- ☐ M_{lower} would be smaller and the confidence interval would be narrower.
☒ M_{lower} would be larger and the confidence interval would be narrower.
☐ M_{lower} would be smaller and the confidence interval would be wider.
☐ M_{lower} would be larger and the confidence interval would be wider.

- (c) (4 points) What statements are true about the confidence interval that was created? **Select ALL that apply.**

- ☐ 54.5% to 72% of all customers are satisfied with the service.
☐ 99% of re-sampled proportions fall within (0.545, 0.72)
☒ We are 99% confident that 54.5% to 72% of all customers are satisfied with the service.
☒ It is possible that fewer than 54.5% of customers are satisfied with the service.

- (d) (5 points) A different resampling procedure is performed based on the null hypothesis $H_0 : p = 0.5$, where p is the true proportion of satisfied customers. A two-sided p-value is reported as 0.001. What conclusions can we draw with a significance level of $\alpha = 0.01$? **Select ALL that apply.**

- ☒ We have evidence that the proportion of satisfied customers is different from 0.5.
☐ We have evidence that more than half of customers are satisfied.
☐ We fail to reject the null hypothesis.
☐ We accept the null hypothesis.
☒ We reject the null hypothesis.

